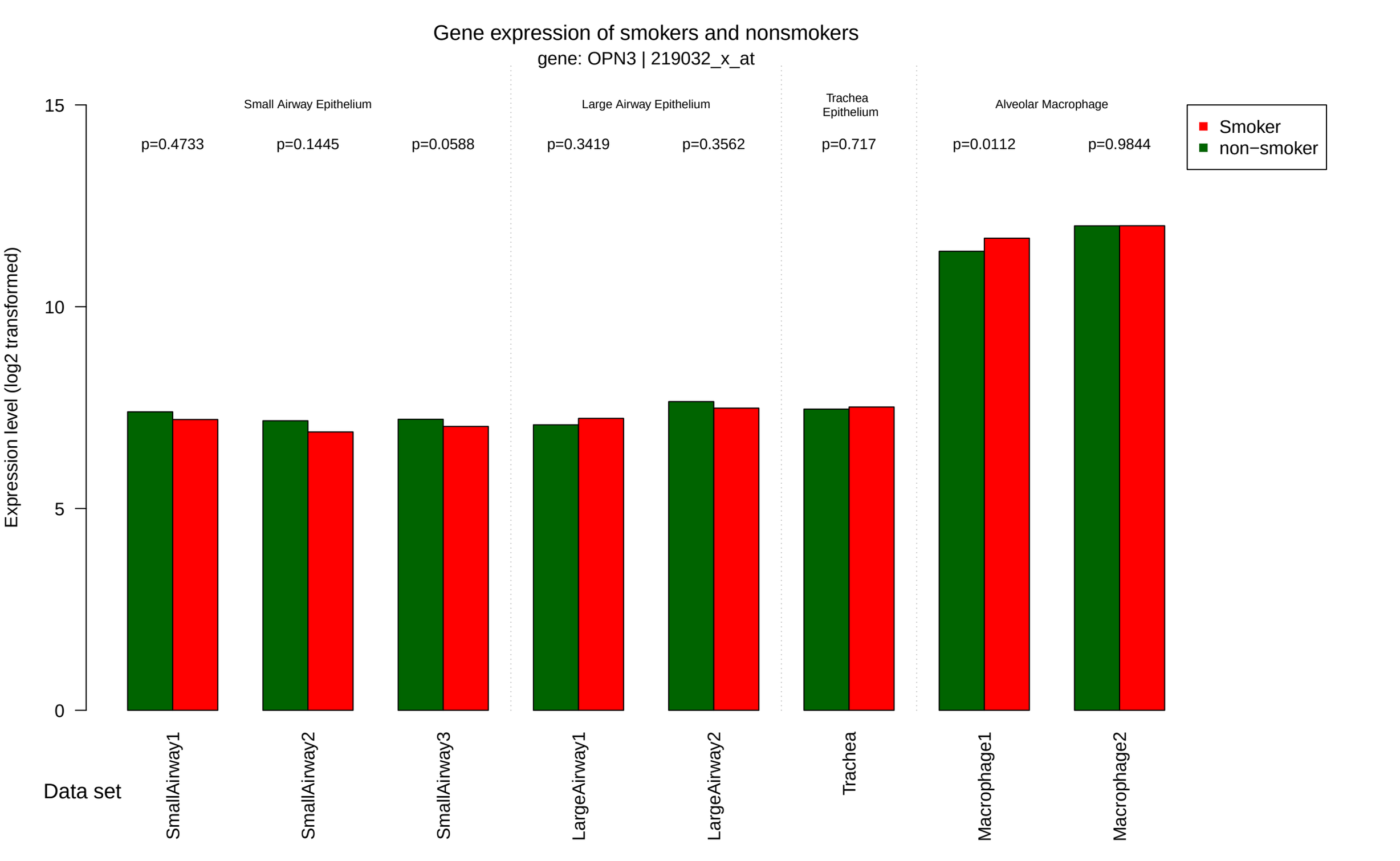
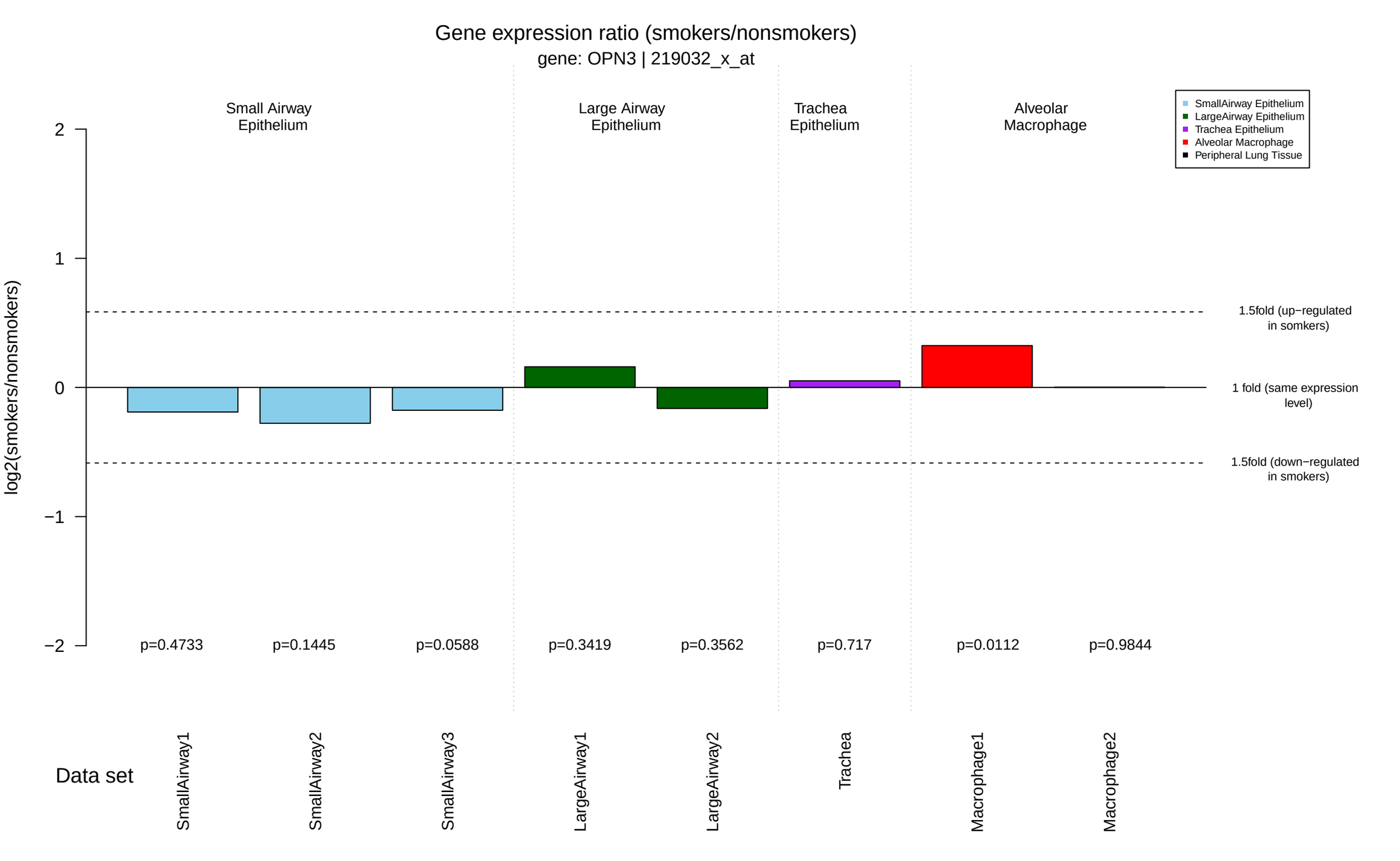
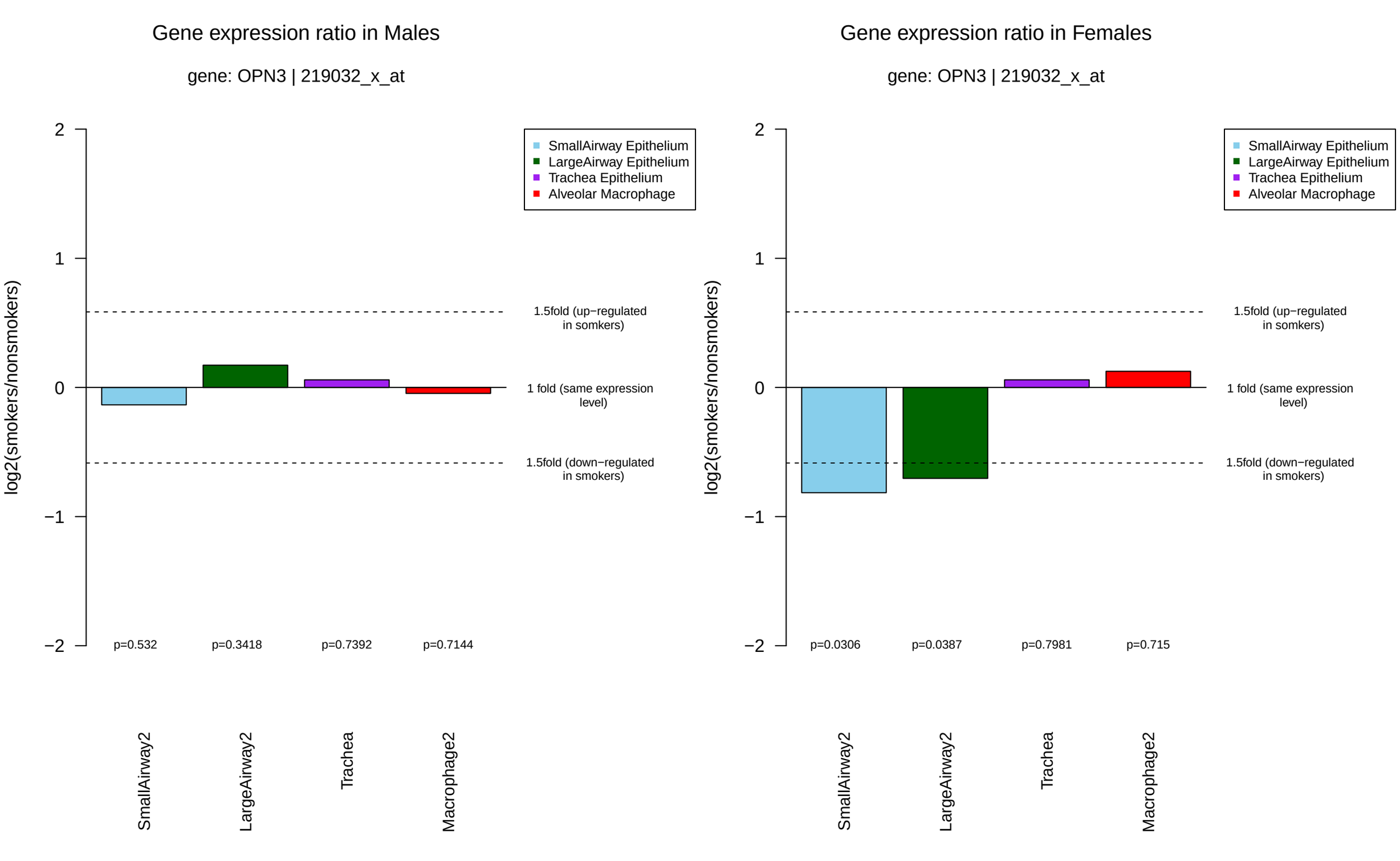
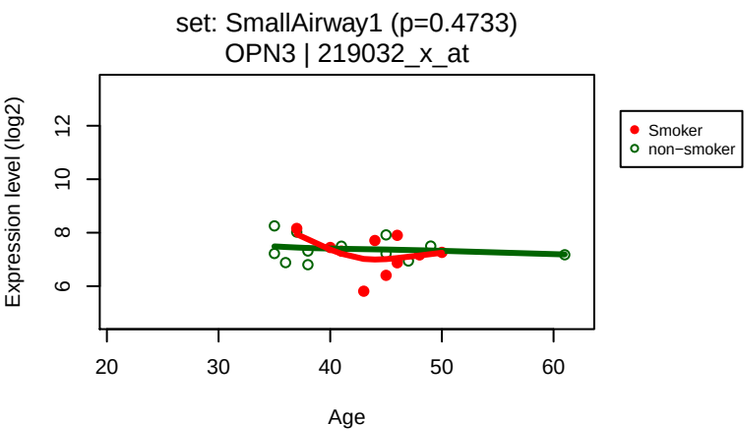
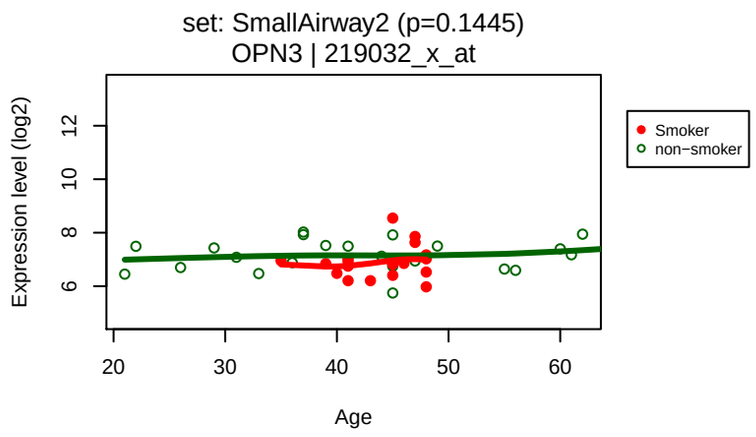
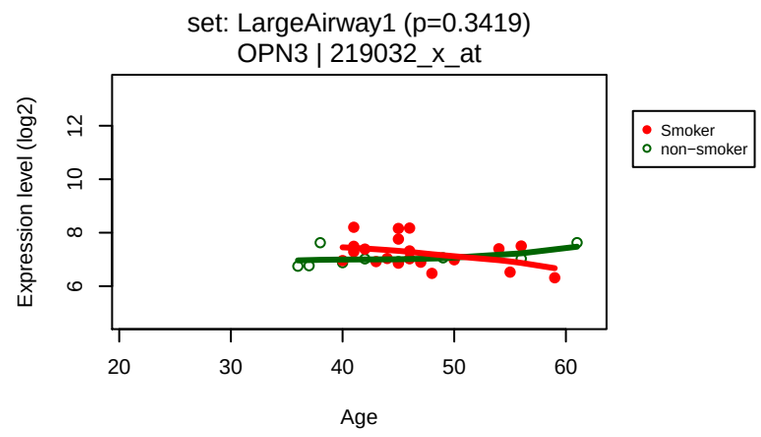

Gene expression of smokers and nonsmokers
gene: OPN3 | 219032_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4733
p=0.1445
p=0.0588
p=0.3419
p=0.3562
p=0.717
p=0.0112
p=0.9844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

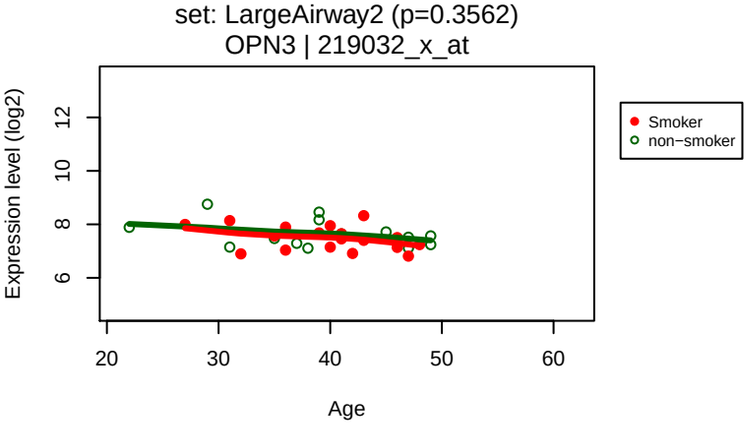

Gene expression ratio (smokers/nonsmokers)
gene: OPN3 | 219032_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4733
p=0.1445
p=0.0588
p=0.3419
p=0.3562
p=0.717
p=0.0112
p=0.9844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

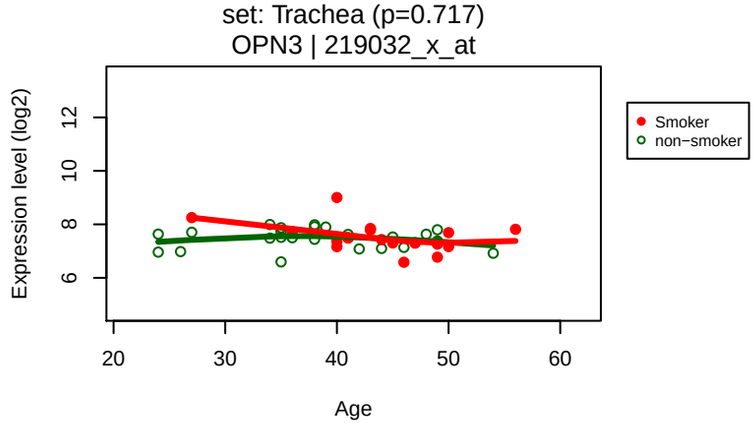

Gene expression ratio in Males
gene: OPN3 | 219032_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.532
p=0.3418
p=0.7392
p=0.7144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPN3 | 219032_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0306
p=0.0387
p=0.7981
p=0.715
SmallAirway2
LargeAirway2
Trachea
Macrophage2

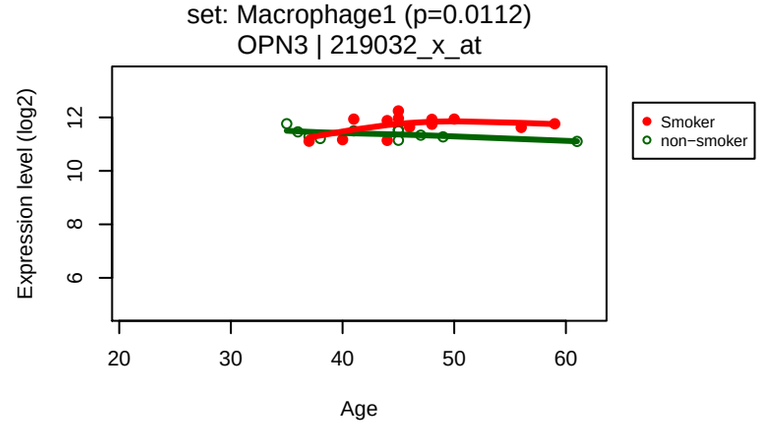

set: SmallAirway1 (p=0.4733)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker

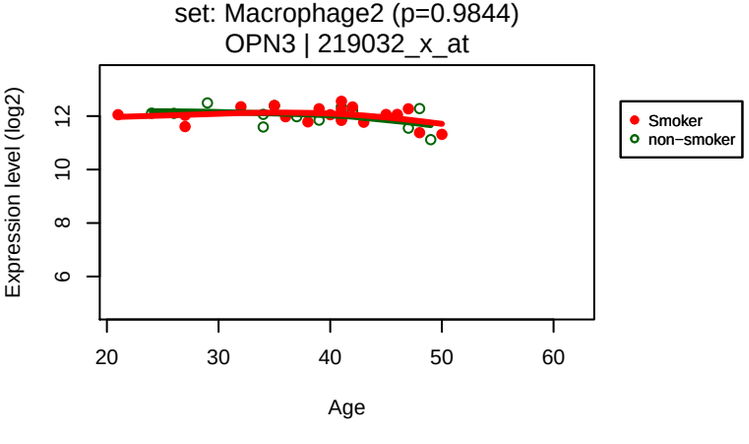

set: SmallAirway2 (p=0.1445)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker

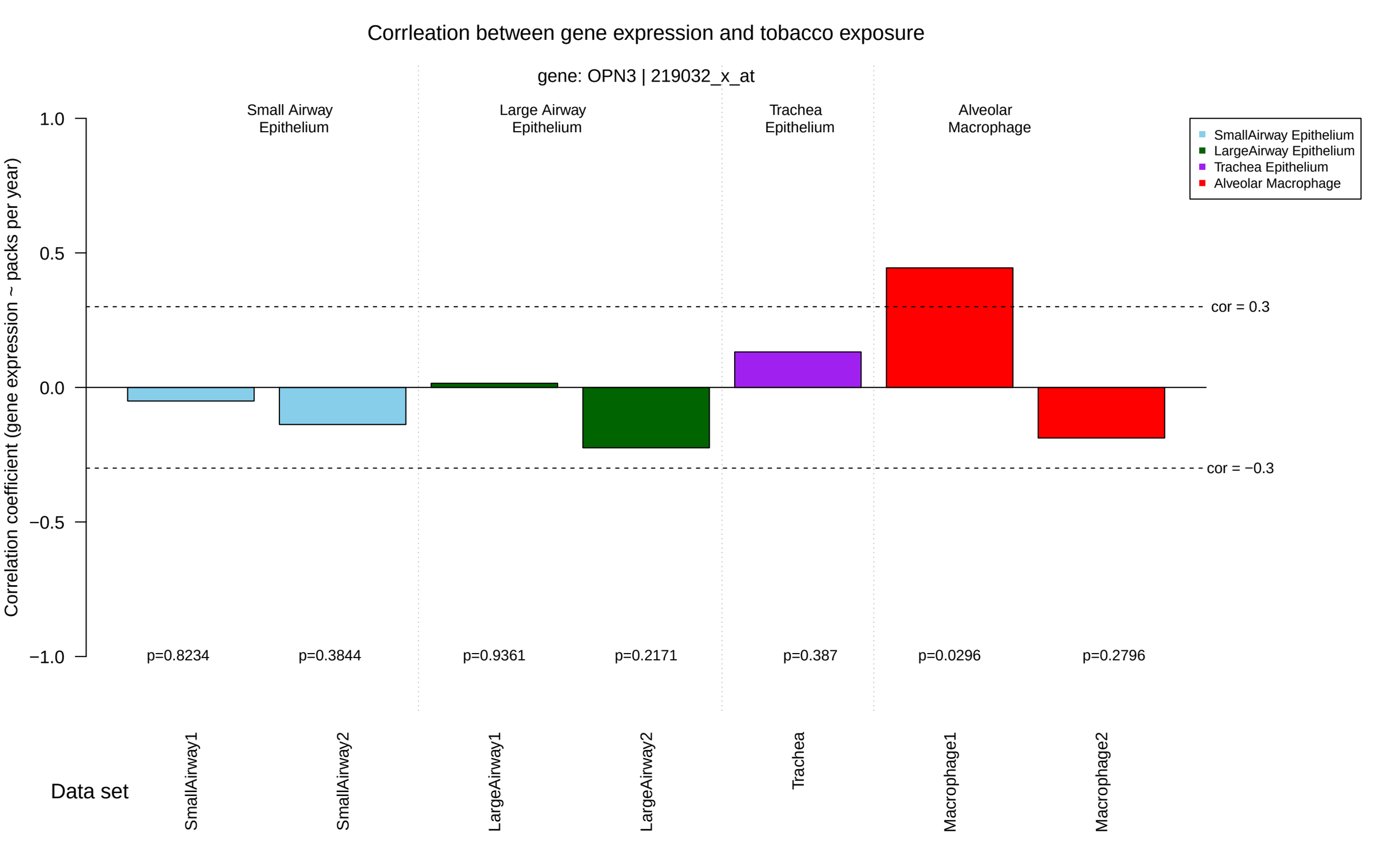

set: LargeAirway1 (p=0.3419)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3562)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.717)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0112)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9844)
OPN3 | 219032_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPN3 | 219032_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8234
p=0.3844
p=0.9361
p=0.2171
p=0.387
p=0.0296
p=0.2796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2